

Large eddy simulation of extreme hydrodynamic forces on structures with mitigation walls using OpenFOAM

Samieh Sarjamee¹ · Ioan Nistor¹ · Abdolmajid Mohammadian¹

Received: 26 October 2016 / Accepted: 1 November 2016 / Published online: 9 November 2016
© Springer Science+Business Media Dordrecht 2016

Abstract The objective of this study is to evaluate the performance of a large eddy simulation (LES) numerical model to assess the influence of mitigation walls with various cross sections on the tsunami-induced loading exerted on a freestanding structure. The validation of the model was performed using laboratory experiment results previously conducted by the second author. The employed experiments simulated a tsunami-like bore at two different impoundment depths, including 55 and 85 cm. The results showed that the one-equation subgrid-scale energy transport LES approach performs better than RANS models such as $k-\epsilon$, $k-\omega$ and LRR and can accurately reproduce experimental data obtained for a wide range of experimental conditions.

Keywords Hydrodynamic loading · Tsunami · Extreme hydrodynamic forces · OpenFOAM · Force–time histories · Mitigation wall · Large eddy simulation · RANS

1 Introduction

The occurrence of natural disasters and associated coastal damage, such as floods and tsunamis, is predicted to increase. Coastal community and infrastructure exposure to risks from coastal hazards such as tsunamis and storm surges are increasing because population

✉ Samieh Sarjamee
ssarj072@uottawa.ca

Ioan Nistor
inistor@uottawa.ca

Abdolmajid Mohammadian
amohamma@uottawa.ca

¹ Department of Civil Engineering, University of Ottawa, 161 Louis Pasteur, Ottawa, ON K1N 6N5, Canada

density and volume in vulnerable coastal areas have grown significantly over the past decades.

Tsunamis are a threat to people's lives and coastal communities. One of the primary mitigation strategies is horizontal evacuation to uphill or inland locations, which requires significant warning time to execute effectively. This has led researchers to turn their attention to improving tsunami warning systems which are designed to increase awareness of an impending tsunami danger and help evacuate the tsunami hazard areas in time. This strategy can be effective to save human lives. However, in case where the affected coastal regions are low-lying, horizontal evacuation does not lessen the huge social, economic, environmental and political losses. In this case, vertical evacuation using tsunami-resistant structures can be an ideal mitigation strategy and, hence, studies are required to fully understand tsunami loading and to determine the optimal structure capable of withstanding the tsunami impact and loading.

While experimental studies can be used to understand different aspects of the phenomena, they are typically expensive, time-consuming and prone to scale effects and measurement inaccuracies. Numerical modeling, on the other hand, is an alternative that, if demonstrated to be accurate enough, could be much less expensive, faster, as well as capable of assessing a wider range of scenarios and also applicable to the actual scale of the problem. This motivates the present study where the objective is to examine various numerical modeling approaches and provide an insight into their performance in the context of tsunami bores and their impacts on coastal structures.

Various coastal structures such as seawalls, bulkheads and breakwaters have been proposed by researchers as protection methods for coastal regions. The main objective of the construction of these structures is to avoid the subsequent inundation and lessen the possibility of further damages by flooding and bores impacts. There are numerous papers on the subject of seawall and beach interaction. Kraus and McDougal (1996) reviewed 40 papers on this subject and found that there is controversy regarding the effectiveness and functionality of seawalls. Some people believe that seawalls are not only costly and intrusive on shoreline views, but that they also limit beach access and interrupt longshore sediment transport. In addition, seawalls have also failed to fulfill expectations of protection against tsunamis. In the past, seawalls have been toppled and tsunami-induced bores have displaced, overtopped and destroyed many of the seawalls built to protect against such phenomena. For example, in 1993 at Okushiri Island, Japan, a tsunami caused 185 deaths in spite of a 4.5-m-high seawall that was constructed to shield this area. An investigation conducted on the post-tsunami surviving buildings showed that there were some factors which played a major role in reducing the tsunami-induced force. These factors include the orientation and design of the buildings as well as the existence of low-height walls located upstream of the structures, which proved to be effective at reducing the exerted tsunami force by dissipating their energy (Lukkunaprasit and Ruangrassamee 2008). Nowadays, researchers are trying to understand the characteristics of tsunami-induced forces in order to design tsunami-resistant buildings or structures.

In order to validate the performance of different RANS and LES approaches, this paper compares the results of multiple numerical experiments performed to study the influence of mitigation walls with various shapes and locations on reducing the force exerted on a freestanding square-shaped column. In this paper, a computational model of the laboratory experiments conducted by Al-Faesly et al. (2012) is developed, and the numerical results are analyzed in comparison with the physical test data in order to validate the model.

This paper is organized as follows. Section 2 provides a literature review on the subject investigated, while the numerical model is described in Sect. 3. Section 4 deals with

structural and walls geometry and location as well as the initial boundary conditions. In Sect. 5, comparisons of the numerical results with experimental data are presented for various cases and the performance of various numerical models is discussed and evaluated. Finally, some concluding remarks complete the study.

2 Literature review

There have been several studies investigating the shear force exerted on structural elements such as vertical walls. However, few studies have directly focused on the efficiency and effectivity of mitigation walls in reducing the force generated by the impact of a tsunami bore on near-shore structures. To the best knowledge of the authors, LES modeling of tsunami bores using an equation for subgrid-scale kinetic energy has not been performed prior to this study. Furthermore, numerical simulation of the effect of low-height mitigation walls on force reduction for tsunami bores has not been conducted before as well.

Investigations on tsunami loading began with the experimental tests conducted by Stoker (1957), aiming to investigate tsunami loading on a vertical wall. He concluded his study by proposing an equation for estimating tsunami loads. Cumberbatch (1960) attempted to estimate the force imposed on a rigid wall originating from a water wedge which was assumed to be two-dimensional before impact. His two proposed solutions could not describe the flow on all regions located between the tip and the origin and was only valid for regions far from and close to the contact point. Hence, he proposed matching the solutions so that his formula could cover all the regions. Laboratory tests conducted by Fukui et al. (1963) focused on estimating the base shear force exerted on dikes by applying two different tank sizes to determine how different scales could affect the results. Cross (1967) tried to refine Cumberbatch's (1960) proposed formula for the shock pressure of a small-scale tsunami wave on a vertical wall by adding a hydraulic term. Ramsden and Raichlen (1990) also focused on the interaction of a tsunami force and a structural element or a vertical column. Their findings revealed that the maximum run-up force was responsible for imposing the maximum force on the wall. A similar experiment was also performed by Arikawa (2008) on a vertical concrete wall.

As mentioned above, the earliest studies on hydraulic bore-induced forces focused on structural elements which later switched to 3D structures. Extensive physical and numerical studies have been performed to date. Arnason's (2005) PhD dissertation objective was to determine how tsunami forces could affect onshore structures. Asakura et al. (2002), Fujima et al. (2009) and FEMA p-646 (2012) proposed formulae for estimating tsunami-induced forces on structures. These formulae were an improvement in comparison with the previous ones but were not, however, capable of covering all the effective parameters such as the transitory characteristic buoyancy forces. Thusyanthan and Madabhushi (2008) and Cox et al. (2010) examined the tsunami loads on a model of a wood structure coastal house. In a similar experiment performed by Bandara and Dias (2012), a real two-story school building similar to those that withstood the 2004 Indian Ocean Tsunami was used to study different aspects of tsunami loads. The results obtained in these types of studies are case dependent, and they are hence limited playing a significant role in the design process yet.

Various studies have been performed to numerically examine the effect of tsunami-induced bores on structures. Lagrangian-based particle methods have been used by researchers to study the effect of tsunami waves on structures. Gotoh et al. (2004) applied a

combination of SPH (smoothed particle hydrodynamics) and LES models to study the tsunami interaction with partially immersed breakwaters. Gotoh et al. (2005) simulated the overtopping process over vertical seawalls. St-Germain et al. (2013) studied numerically and experimentally the influence of tsunami bores on columns of square cross section by employing a three-dimensional smoothed particle method numerical method. The item in which a significant difference was observed between the collected data and simulation results was the presence of an impulsive force predicted by the SPH model but not observed in the physical test. Liu et al. (2015) also applied a SPH model to study the propagation of tidal bores numerically and compared their results with the similar experimental data performed. The comparison showed a good agreement between numerical and physical results. Huang and Zhu (2015) proposed a moving particle simulation (MPS) approach which is also a Lagrangian approach method. It does not need any computational mesh and is able to accurately simulate the interaction between tsunami waves and structures.

Nouri et al. (2010) conducted an experimental test program to study the influence of tsunami bores in laboratory in terms of the exerted force and pressure on the structure. They considered two freestanding structures with circular and square cross sections located close to the downstream end of a flume with high discharge. The forces on the structure were recorded as well as the pressure distribution exerted on them generated by the bore. A good agreement was observed with the results of similar experiments available in the literature at the time. However, some discrepancies with tests of Árnason (2005) were observed in the pressure–time histories of the structure (the upstream side).

The recent experimental work of Al-Faesly et al. (2012) is most closely related to this study. The authors of this comprehensive experimental program investigated the use of a low-height mitigation walls located upstream of the structure (Fig. 1).

Such walls are expected to have a damping effect on the base shear force and can lessen the effect of the tsunami-driven debris field. The authors focused on the effect of flat and curved mitigation walls while varying their height and location on the exerted force on the square column. They found a relationship between the walls parameters and the force exerted by the tsunami-like bore on freestanding structures. The results revealed that, if located correctly, using protection walls with various cross sections can be a mitigation strategy to decrease the exerted overall shear force on the column. In terms of the flat

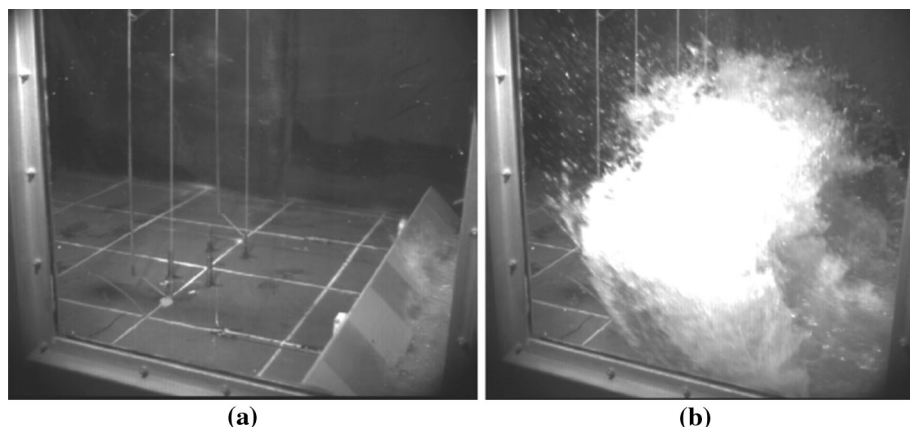


Fig. 1 Bore front impact and deflection by the 45° mitigation wall (Al-Faesly et al. 2012)

mitigation wall with angles of 45° and 90° relative to the bottom of the flume and placed at the distance equal to the cross-sectional dimension of the square model (1D), the magnitude of the base shear force decreased considerably. Extending the location of the protection wall from 1D to 3D (at a distance three times the width of the column) was also effective but had a less pronounced effect in reducing the base shear force—this was valid only for the test with the vertical mitigation wall. For the test with an inclined wall, it was not only ineffective but also harmful. In terms of curved mitigation wall, the results showed the most effective location for this mitigation wall was 3D. The location 1D, which was closer to the structure, resulted in a higher magnitude of base shear force on the column.

Rahman et al. (2014) conducted an experimental study to investigate the force reduction effects of seawalls on onshore structures. He focused on two parameters, including the location and height of the wall, to determine their effect on the generated force. Based on his findings, he concluded that seawalls located close to the coastal structure with adequate height are capable of significantly reducing the exerted base shear force. The results also revealed that the efficiency of a porous wall is equal to the efficiency of a solid wall. A large-scale experimental study of Oshnack et al. (2009) using a large-scale flume located at Oregon State University evaluated the effectiveness of a small seawall in reducing the exerted tsunami load. His results indicated that the small seawall is effective in reducing the wave force due to the absence of spike in the force–time histories. The author concluded that small seawalls can also be useful in decreasing the impulsive force. Thomas and Cox (2011) proposed an equation capable of estimating the tsunami force reduction due to a seawall.

A reconnaissance visit performed by a Japanese team of researchers from the Research Center for Disaster Reduction Systems (RCDRS 2005) and surveys on Phuket, Thailand, after the 2004 Indian Ocean Tsunami revealed that coastal geomorphic features, such as sand dune belts, associated forests, mangroves and low-height seawalls, played a significant role in reducing the destruction due to the tsunami waves by decreasing the bore velocity. For example, the Kata–Karon area of Phuket reported less damage due to the fact that some buildings were constructed behind a sand dune. It is interesting to note that the tsunami bore overtopped the sand dunes, such that while the damage due to direct flooding was inevitable, the damage due to the direct tsunami bore impact was omitted. Another example is Patong Beach, where no serious structural damage was reported. The low seawall acted as tsunami bore deflector and, as a result, a considerable reduction in the magnitude of the exerted force on the impacted structures was noticed. The field survey indicated that seawalls made of sandbags or landward-sloping seawalls are less effective as coastal protection structures.

3 Development of the numerical model

The software used to develop this newly proposed numerical model is OpenFOAM v2.3.0 which runs on the Linux operating system. OpenFOAM is a free, open-source software package capable of solving problems of continuum mechanics such as computational fluid dynamic (CFD) analysis. There are several models and solvers written in C++ language available in OpenFOAM which are applicable for various engineering and science problems. There are various tools for meshing, preprocessing and post-processing, and one of the most interesting functional aspects of this software is its open-source feature which

enables the users to customize its existing functionality. Having access to the source code is a powerful feature which offers those who have a background in programming the possibility to assess and modify the codes. It is worth noting that the programming language of the available libraries in OpenFOAM is object-oriented, which provides programmers with a new approach to overcome the complexity of traditional programming languages. OpenFOAM has good extensibility qualities. For instance, users are able to create custom objects such as boundary conditions and incorporate turbulence models which existing solvers can respond to. Therefore, there is no need to change and recompile the source code. This software is capable of simulating a broad range of engineering problems by applying the finite volume method (FVM) that is inherently mass conservative (OpenFOAM 2014).

The OpenFOAM solver applied for this case is InterFoam. It is capable of handling the problems with free surface where two fluids are separated from each other through a sharp boundary. To describe the flow, the procedure of discretization is performed by using a finite volume technique, the applicable equations are incompressible RANS, and the turbulence is modeled by applying an LES model (large eddy simulation) as well as the SGS are modeled by applying a one-equation subgrid-scale energy model, which is explained later herein. Structured grids are applied to the simulation domain in this study (OpenFOAM 2014).

3.1 Governing equations

The Navier–Stokes equations are the principal partial differential equations capable of predicting the motion of fluid by describing how flow parameters such as pressure, density, velocity and temperature are related and are derived from conservation laws. These parameters can be obtained directly (DNS), in temporal averaged form (Reynolds-averaged Navier–Stokes, RANS) or by using the spatially filtered form (LES) as described in the following section.

3.1.1 Reynolds-averaged equations

Using tensor notation, the differential form of the continuity equation for a three-dimensional system is written as (Chamani et al. 2013, p. 209):

$$\frac{\partial \bar{u}}{\partial x} + \frac{\partial \bar{v}}{\partial y} + \frac{\partial \bar{w}}{\partial z} = 0 \quad (1)$$

where $\bar{u}$, $\bar{v}$, and $\bar{w}$ are the mean velocity components and $\acute{u}$, $\acute{v}$, $\acute{w}$, and $\acute{p}$ are instant velocity in the x , y and z directions, respectively:

$$u(t) = \bar{u} + \acute{u} \quad v(t) = \bar{v} + \acute{v} \quad w(t) = \bar{w} + \acute{w} \quad p(t) = \bar{p} + \acute{p} \quad (2)$$

The Reynolds-averaged continuity equation is based on the average flow velocity.

Similarly, the differential form of the momentum equation for a three-dimensional system can be written as (Chamani et al. 2013):

$$\bar{u} \frac{\partial \bar{u}}{\partial x} + \bar{v} \frac{\partial \bar{u}}{\partial y} + \bar{w} \frac{\partial \bar{u}}{\partial z} = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial x} + g_x + \nu \left(\frac{\partial^2 \bar{u}}{\partial x^2} + \frac{\partial^2 \bar{u}}{\partial y^2} + \frac{\partial^2 \bar{u}}{\partial z^2} \right) - \left(\overline{\acute{u}^2} + \overline{\acute{u}\acute{v}} + \overline{\acute{u}\acute{w}} \right) \quad (3)$$

$$\bar{u} \frac{\partial \bar{v}}{\partial x} + \bar{v} \frac{\partial \bar{v}}{\partial y} + \bar{w} \frac{\partial \bar{v}}{\partial z} = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial y} + g_y + \nu \left(\frac{\partial^2 \bar{v}}{\partial x^2} + \frac{\partial^2 \bar{v}}{\partial y^2} + \frac{\partial^2 \bar{v}}{\partial z^2} \right) - \left(\frac{\partial \overline{v^2}}{\partial y} + \frac{\partial \overline{uv}}{\partial x} + \frac{\partial \overline{vw}}{\partial z} \right) \quad (4)$$

$$\bar{u} \frac{\partial \bar{w}}{\partial x} + \bar{v} \frac{\partial \bar{w}}{\partial y} + \bar{w} \frac{\partial \bar{w}}{\partial z} = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial z} + g_z + \nu \left(\frac{\partial^2 \bar{w}}{\partial x^2} + \frac{\partial^2 \bar{w}}{\partial y^2} + \frac{\partial^2 \bar{w}}{\partial z^2} \right) - \left(\frac{\partial \overline{w^2}}{\partial z} + \frac{\partial \overline{vw}}{\partial y} + \frac{\partial \overline{uw}}{\partial x} \right) \quad (5)$$

where ρ is the fluid density, p is the fluid pressure, t is the time, and g is the gravitational acceleration.

3.1.2 Large eddy simulation model (LES)

There is a wide variety of turbulence models which can be used to simulate turbulence. Determining the most suitable model depends on the nature of the problem since no model provides a fully satisfactory solution. One modeling approach, initially proposed in 1963 by Joseph Smagorinsky, is LES, which is proven to be more accurate than Reynolds-averaged Navier–Stokes equations (RANS) and computationally less expensive than direct numerical simulation (DNS). The main drawback of the RANS model is the fact that this model is responsible for representing a broad range of scales, which could ultimately affect the accuracy of the results. Small scales and large scales are dependent on viscosity and boundary conditions, respectively, and as a result, it does not seem to be possible to consider the same approach for modeling the effect of large scales for different flows. The principal shortcoming of DNS is the fact that the precise result is highly dependent on very fine grid points to capture all the scales of the motion, which is computationally very expensive. As a result, this model is more applicable to simple geometries at low Reynolds numbers. LES is an intermediate method between DNS and RANS and has become one of the most successful models for turbulence. This model's principal operation is based on resolving large scales while modeling the small subgrid scales. In comparison with RANS, LES is more precise because it does not cover the entire spectrum of length scales and it models only the small scales of the flow. As a result, it is an appropriate approach for turbulence at high Reynolds numbers. Furthermore, LES operates through applying a spatial filtering approach via dividing the large eddies which are resolved from the small eddies that are modeled. A proposed convolution for filtering process is Piomelli (1999).

$$\bar{\phi}(\zeta) = \int_D \phi(\zeta') G(x, \zeta'; \bar{\Delta}) d\zeta' \quad (6)$$

where G is the spatial filter function, $\bar{\Delta}$ is the filter width, and D is the domain of integration. The role of the filter function is to determine the magnitude of the small scale.

LES models use the Navier–Stokes equations. Hence, the governing equations are filtered N–S system. The filtered governing equations are expressed as follows (Piomelli 1999).

$$\frac{\partial \bar{u}_i}{\partial x_i} = 0 \quad (7)$$

$$\frac{\partial \bar{u}_i}{\partial t} + \frac{\partial}{\partial x_j} (\bar{u}_i \bar{u}_j) = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial x_i} - \frac{\partial \tau_{ij}}{\partial x_j} + \nu \frac{\partial^2 \bar{u}_i}{\partial x_j \partial x_j} + g_i \quad (8)$$

where $\bar{u}_i$ is the velocity component in the x_i direction, $\bar{u}_i$ is the resolved velocity component in the x_i direction, p is the pressure, ρ is the fluid density, and τ_{ij} is the subgrid-scale stress tensor. Equations (7) and (8) are widely used to model the large-scale flow, while for the small scales of the flow, a subgrid-scale stress modeling term (SGS) is required because some scales of flow, namely the dissipative scales, are not resolved in LES. As a result, subgrid-scale modeling is employed to dissipate energy:

$$\tau_{ij} = \bar{u_i u_j} - \bar{u_i} \bar{u_j} \quad (9)$$

There is a broad range of proposed SGS models in the literature. A popular group is based on the eddy-viscosity hypothesis and equilibrium assumption, which states that equilibrium condition is dominant for small scales and their main role is dissipating the energy they receive from the resolved scales

$$\tau_{ij} - \frac{1}{3} \delta_{ij} \tau_{kk} = 2\mu_t \bar{s}_{ij} \quad (10)$$

where τ_{ij} is the subgrid-scale stress tensor, μ_t is the turbulent viscosity, δ_{ij} is the Kronecker tensor, $\bar{s}_{ij}$ is the resolved strain rate tensor. A one-equation model is employed in this study to account for the SGS energy dissipation. The role of this model is to estimate the velocity scale by solving an equation for the transport of subgrid-scale energy. This equation can be expressed as follows:

$$\frac{\partial K_{\text{sgs}}}{\partial t} + \frac{\partial}{\partial x_j} (K_{\text{sgs}} \bar{u}_j) = + \frac{\partial}{\partial x_j} \left[\left(C_k K_{\text{sgs}}^{\frac{1}{2}} \bar{\Delta} + \nu \right) \frac{\partial K_{\text{sgs}}}{\partial x_j} + \tau_{ij} \bar{u}_i \right] - C_\epsilon \frac{K_{\text{sgs}}^{\frac{3}{2}}}{\bar{\Delta}} - \tau_{ij} \bar{s}_{ij} \quad (11)$$

and

$$\nu_T = C_\nu K_{\text{sgs}}^{1/2} \bar{\Delta} \quad (12)$$

where K_{sgs} is the subgrid-scale energy that is equivalent to $\tau_{kk}/2$, C_k and C_ϵ are SGS stress model coefficients, $\bar{\Delta}$ is the principal (grid) filter width, $\bar{u}_j$ is the velocity component in the x_j direction, ν is the kinematic viscosity, and $\bar{s}_{ij}$ is the resolved strain rate tensor (Piomelli 1999).

3.2 Treatment of the free surface

One proposed interface-capturing method capable of tracking the free surface is the volume of fluid (VOF) technique. This method calculates the flow occurring in a two-phase domain, such as air and water in this study, by considering each phase as a single fluid that experiences variation of physical properties through the interface. In order to determine the value for variables such as density ρ and the value of the volume fraction c (Andrillon and Alessandrini 2004) should be computed first because the values of ρ and μ are dependent on that parameter: c can take values of 0 or 1 for the air and water phase, respectively. For a system with two phases, the VOF method considers an advection process for the volume fraction. The role of advection term is to determine the location of the free surface, and its main advantage is that it provides sharp boundaries for the free surface, which is sensitive to the grid size in most other methods.

The transport equation employed in the VOF method is written as follows (Heyns et al. 2013),

$$\rho = c\rho_1 + (1 - c)\rho_2 \quad (13)$$

$$\frac{\partial c}{\partial t} + \vec{\nabla} \cdot (c\vec{u}_c) = 0 \quad (14)$$

where $\vec{u}_c$ can be defined as the estimated relative velocity between air phase and water phase.

3.3 Computational time step

Determining an appropriate time step is a key and challenging aspect of any computational simulation. While a shorter time step can lead to more accurate results, it inherently increases the simulation time. Adaptive time stepping is a technique which allows varying the time step as the estimation proceeds, rather than defining a constant value for the step size for the entire calculation. This technique exists in the finite volume library of OpenFOAM, and it is applied in some standard solvers such as InterFoam. It is based on keeping a certain max Courant number which is defined below (Berberović 2010):

$$Co = \frac{|U_f|}{d} \Delta t \quad (15)$$

where Δt is the time step and the vector d is the distance between the interfaces of two computational cells. The following formula can be used for estimating the value of the Courant number (Co_0) in order to estimate the time step.

$$\Delta t = \min \left\{ \min \left[\frac{Co_{\max}}{Co_0} \Delta t_0, \left(1 + \lambda_1 \frac{Co_{\max}}{Co_0} \right) \Delta t_0 \right], \lambda_2 \Delta t_0 \right\}, \Delta t_{\max} \quad (16)$$

where $Co_{\max}$ is the maximum courant number specified by the user. The role of the two variables $\lambda_1 = 0.1$ and $\lambda_2 = 1.2$, applied in the above equation, is to prevent numerical oscillations by controlling the increase in the time step.

4 Domain geometry and initial and boundary conditions

The numerical experiments are designed to reproduce the laboratory experiments conducted in a flume with a reservoir containing impounding water at rest, located at the upstream side of a column which has a square cross section (0.305 m by 0.305 m) and a height of 1 m. The dimensions of the flume including its length, width and height were 13.17, 2.7 and 1.4 m, respectively. The speed of the bore tip at the upstream side of the structure before hitting and without mitigation wall was determined to be 2.37 m/s. The impoundment water depth was 55 cm for the case of curved mitigation wall and 85 cm for the case of inclined mitigation wall. The laboratory experiment was conducted at the National Research Council (NRC), Canada (Al-Faesy et al. 2012). Due to the demonstrated resemblance between tsunami-induced bores and the generated bore in a dam break case (Chanson 2006), the hydraulic bore generated in their study is produced by a sudden water release. Two different shapes of mitigation walls in the form of inclined and curved walls were placed at two locations upstream of the column. The objective of the experiments was to examine whether the low-height wall could be effective in reducing the

tsunami hydrodynamic force and the associated initial spike induced by the tsunami bore on the column. The tests were also repeated without any mitigation wall in the flume in order to compare the data with the condition where the mitigation wall is installed. The rapidly opening swing gate was opened, and the impoundment water was released inside the flume in the form of a dambreak wave. The water impacted first the mitigation wall before impacting the column and, as a result, it produced a complicated flow pattern. At time = 0, the initial velocity, pressure and phase fractions for all the domain computational cells except for the reservoir were set equal to zero as the experiments were conducted over dry bed (that is, the flume is initially dry before opening the gate of the reservoir). Phase fraction was set to 1 for the reservoir cells (Fig. 2).

Flume geometry (Fig. 3) is modeled to exactly reproduce the National Research Council (NRC) flume in order to allow for model validation and for an accurate comparison of the numerical and experimental results.

In the numerical model, the boundary condition of all of the vertical (lateral) and horizontal (bottom) faces of the L shape flume was defined as solid walls, except for the outlet at the downstream of the flume, which was considered as a zero gradient condition in order to directly evacuate flow out and thus avoid water accumulation at the outlet. Velocity at the computational cells of the wall was set to zero. Several structured grids with the cell dimension of $2 \times 2 \times 2$ cm were generated for each case to numerically replicate the physical model.

In this numerical study, the wall function method was applied to account for viscous effects on the transport processes at the wall. In order to compute turbulent viscosity (ν_T), boundary conditions for ν_T have to be defined. OpenFOAM offers various ν_T wall functions, including `nutKWallFunction`, which was used in this study as a generic wall function suitable for a high-Re turbulence models. This boundary condition provides a turbulent kinematic viscosity condition when applying wall functions, based on turbulence kinetic energy.

5 Model validation

The validation of the proposed numerical model was performed by comparing experimental and simulated results for several case studies. It should be noted that, in order to ensure the model is independent of mesh resolution, a sensitivity analysis was performed to

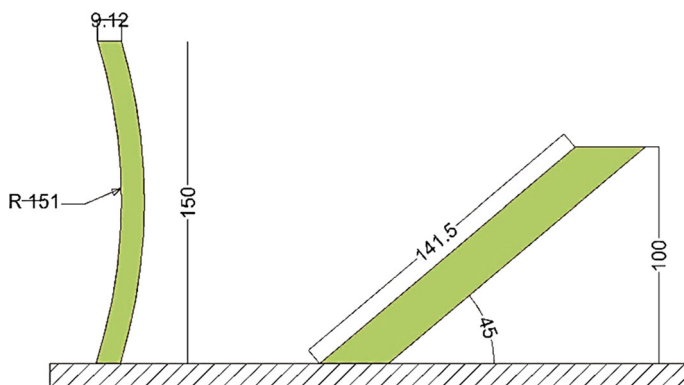


Fig. 2 Schematic of the mitigation wall cross sections (dimensions in mm). *Left* Curved wall, *Right* 45° inclined wall

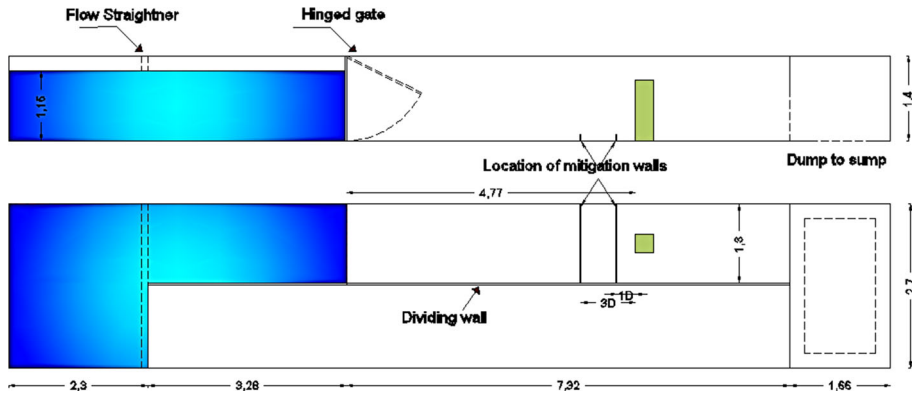


Fig. 3 Side (top) and plan (bottom) views of the NRC flume with mitigation walls located at 1D and 3D (dimensions in m)

establish the suitable grid for the proposed model. This analysis was performed by running several simulations with different grid sizes starting at 3 cm, then 2 cm and, finally, 1 cm. The 2-cm grid size yielded the most accurate results, while the 1-cm grid size did not significantly improve them. Thus, all the simulations are performed using the 2-cm grid size.

In order to develop an accurate model capable of simulating these cases accurately, various models such as RANS ($k-\epsilon$, $k-\omega$ and LRR) and LES were tested. Figure 4 depicts the total hydrodynamic force for dry bed condition with curved mitigation wall located at 1D and an initial impoundment water depth of 55 cm for the simulations that used the LES and RANS ($k-\omega$, $k-\epsilon$ and LRR) turbulent models. The results shown in Fig. 4 revealed that RANS models were not able to simulate these cases accurately. However, the experimental data were accurately reproduced by applying the large eddy simulation approach described in Sect. 3.1.2. Due to the nature of the problems considered in this paper that correspond to turbulence with high Reynolds numbers, LES can be more efficient because it does not cover the entire spectrum of length scales; it directly simulates larger scales of the flow and models only the effect of smaller-scale eddies. Therefore, LES is found to be a more appropriate approach for tsunami bores which are turbulent with high Reynolds numbers as can be observed in Fig. 4. Based on this conclusion, the LES approach is employed in the rest of the paper.

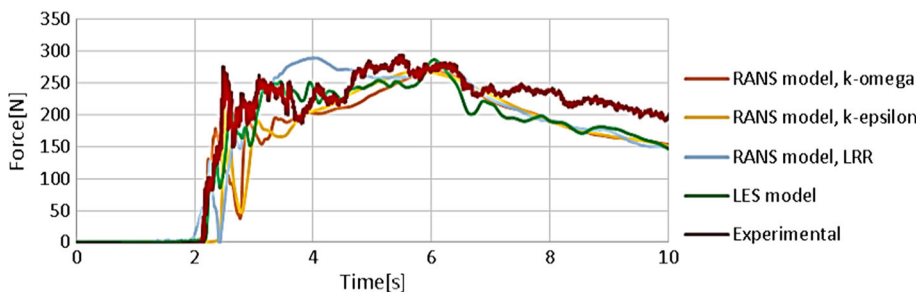


Fig. 4 Examining the effect of various turbulence models

The following five cases were investigated to obtain a better insight into the effects of the mitigation walls in reducing the tsunami-induced force and to examine the capability of RANS and LES models in simulating such cases. The first three cases include examining the influence of curved mitigation walls on the tsunami bore force on the column under the three conditions (1) without the mitigation wall, (2) with the mitigation wall placed at 1D upstream of the column and (3) the wall placed at 3D (where D is the side length of the square column cross section which is of 305 mm). The initial impoundment depth for all three cases was set to 55 cm, and the effect of inclined mitigation wall was examined. The impoundment depth of 85 cm was applied for the last two cases.

5.1 Test Case 1: Effect of bore-induced force on a square cross section column without mitigation walls, water depth 55 cm

In this case, the total net force exerted on the square column without the presence of the mitigation wall was estimated. The base shear force was calculated by integrating the estimated pressure over the entire surfaces of the structure at each time step. The results are shown in Fig. 5.

The initial force spike, which occurred at the beginning of the force–time history, indicates the initial impingement due to the impact of the dambreak wave on the structure. The force data recorded from the experimental test which are also shown in Fig. 5 revealed a peak occurring at $t = 2$ s. This was generated as a result of the initial impact of the rapidly advancing bore across the upstream side of the column. The magnitude of this force was accurately reproduced by the numerical model. However, a small time delay was observed, and this could be attributed to the slight differences in the flume roughness of the flume (slightly rougher in the experiment compared to the model).

Additionally, the gradient of the total force in the model was gradual comparing its steeper variation observed in the experimental case. This could be attributed to the approximation of the eddy viscosity and wall shear stress. Following the initial force peak, a noticeable reduction in the total force was observed—this stemmed from the collapse of the rising water back onto on the incoming flow, reducing the contact surface with the structure as shown in Fig. 6. Following this instant, this volume of water that joins the ongoing incoming flow causes a gradual increase in the calculated force up to its maximum

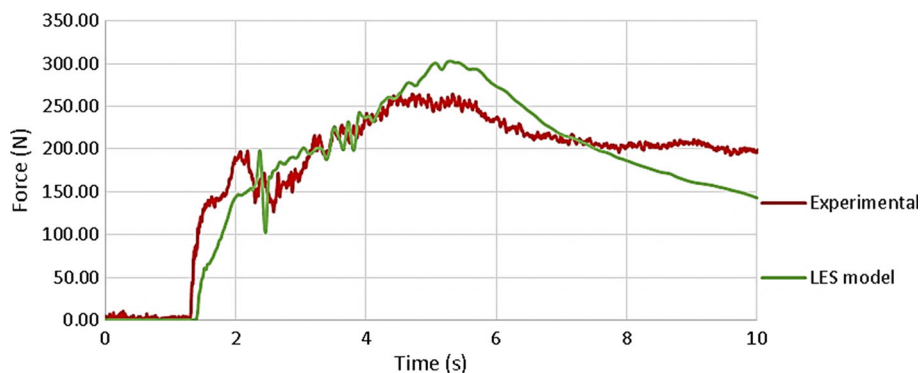


Fig. 5 Test case 1—Total hydrodynamic force for dry bed conditions without mitigation wall. Initial impoundment water depth of 55 cm

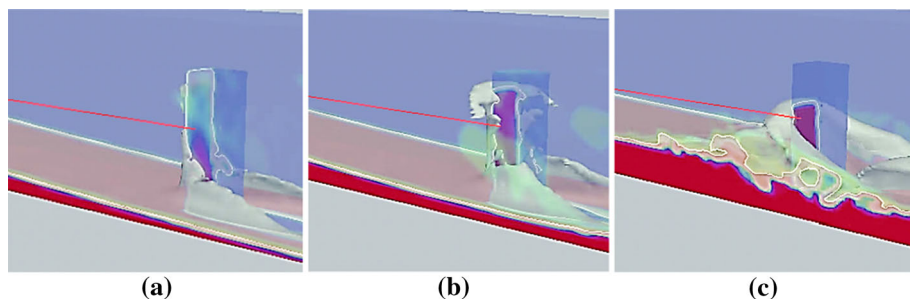


Fig. 6 **a** Initial run-up of the incoming flow onto the upstream side of the structure. **b** Run-up flow begins to return onto the incoming flow. **c** Collapse of the rising run-up of water onto the incoming flow

magnitude peak point. The numerical results are similar to the experimental results up to the maximum run-up force: At this point, a noticeable difference is observed for the magnitude of this maximum force. A similar simulation performed by Douglas and Nistor (2014) using a RANS model revealed divergence between numerical results and experimental data, with different force rise rates at this instant. The magnitude of the maximum force that is due to the run-up was overestimated by about 50 N, and this overestimation was also identified in the study performed by Douglas and Nistor (2015). Finally, the decrease in the calculated net force did not occur at the same rate for the experimental data and numerical results. To the later decreased faster during the last seconds of the simulation due to the fact that the pump used to supply water to the flume continued to operate and, hence, some water continued to flow in the flume—this was neglected in the numerical simulation.

5.2 Test Case 2: Effect of bore-induced force on a square cross section column with a curved mitigation wall located at 1D

The objective of this experimental test case was to examine the effectiveness and functionality of low-height walls in reducing the tsunami-induced force on a structure. In this test case, a 15-cm-high curved mitigation wall with a width of 1 cm was located at 1D (305 mm from the upstream surface of the structure). More details about the experimental test can be found at Al-Faesly et al. (2012). Figure 7 shows that the LES model is capable of reproducing the experimental data, while the RANS model failed to predict accurately. RANS and LES numerical results are compared in Fig. 4.

When comparing experimental data and numerical results, the greatest discrepancy was observed for the predicted magnitude of the impact force, which revealed an underestimation of the OpenFOAM model in capturing the precise experimental variation, which might be the result of approximations in calculation of the shear stress in the wall and eddy viscosity. Comparing the experimental data and numerical results for Case 1 and 2 revealed that placing the mitigation wall at the 1D distance was not effective in reducing the exerted tsunami force on the model structure which could be the result of the fact that the curved mitigation wall blocked the flow of the water, resulting in water accumulation and higher water levels behind and very close to the upstream side of the column (Fig. 8).

This was followed by the collapse of the accumulated water column behind and very close to the model structure which led to a significant run-up force which lasted

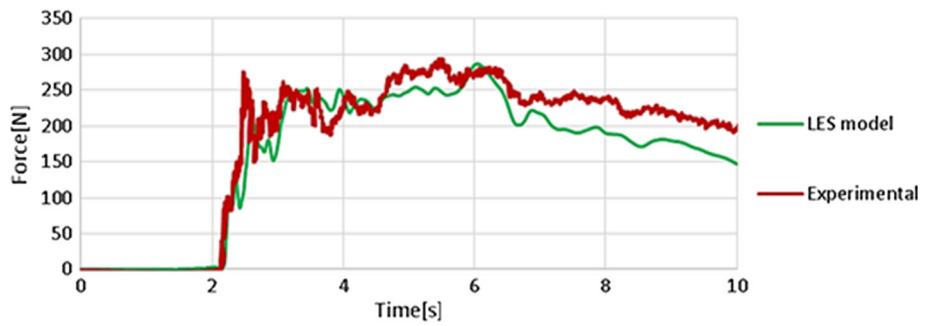


Fig. 7 Test case 2—Total hydrodynamic force for dry bed condition with curved mitigation wall located at 1D and an initial impoundment water depth of 55 cm

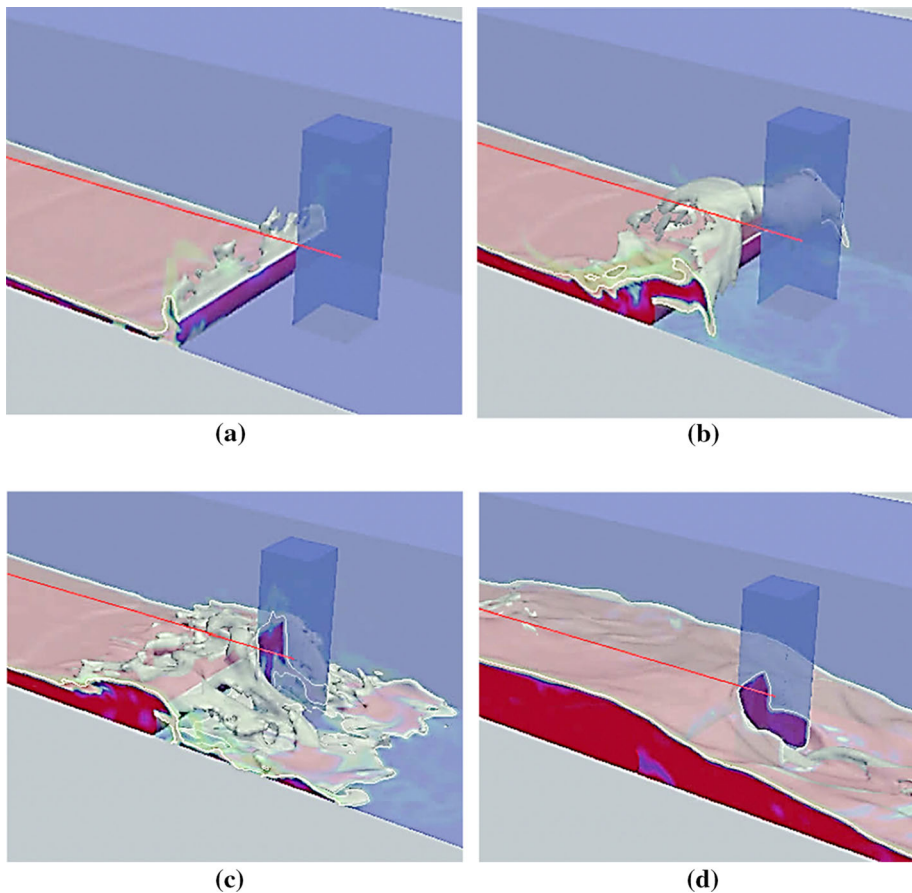


Fig. 8 Water blockage due to the presence of curved mitigation wall; **a** bore front impacts the mitigation wall, **b** incoming flow overtops the mitigation wall, **c** water begins to accumulate behind the mitigation wall and between the mitigation wall and the column, and **d** water accumulates at the upstream side of the column

approximately 3 s. The peak force predicted on the column with mitigation wall at 1D was estimated to be higher in comparison with the case without mitigation wall for the physical test.

5.3 Test Case 3: Effect of tsunami-induced force on a square cross section column with a curved mitigation wall located at 3D

In this test case, the location of the curved mitigation wall was modified from 1D to 3D. The LES model predicted this case very well. Figure 9 shows the comparison between the numerical results and the experimental data.

The largest discrepancy between the experimental data and numerical results occurred at approximately 2.5 s after the initial impact force. At this instant, the predicted model results showed a significant decrease in total force due to the sudden reduction in water level after the initial run-up. Figure 10 reveals that placing the mitigation wall at 3D damped the peak force and impulsive base shear because the location of the impacted water on the column was lower in this case.

In comparison with Test Case 2, the mitigation wall located at 1D is more effective in reducing the dambreak wave-induced force by reducing the speed of the incoming flow and reducing thus the impacted area of the column surface at initial impact.

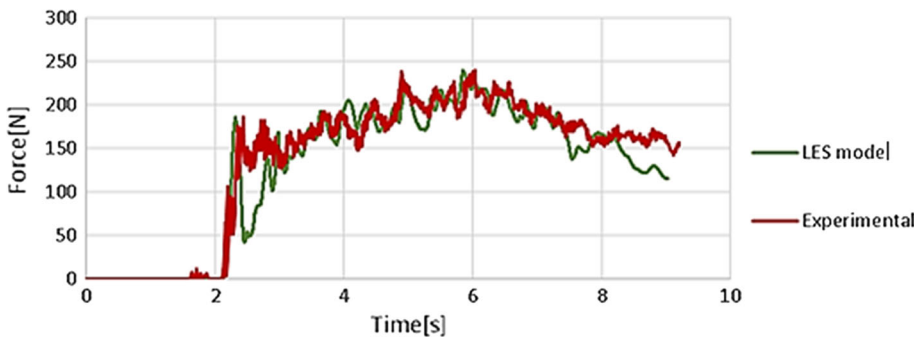


Fig. 9 Test case 3—Total hydrodynamic force for dry bed condition with curved mitigation wall located at 3D and an initial impoundment water depth of 55 cm

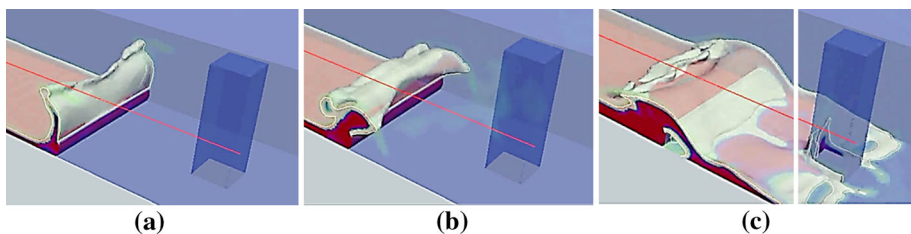


Fig. 10 Water blockage due to the presence of curved mitigation wall; **a** bore front impacts the mitigation wall, **b** incoming flow overtops the mitigation wall, and **c** water accumulates behind the upstream side of the column and reduces the efficiency of the curved mitigation wall located at 3D

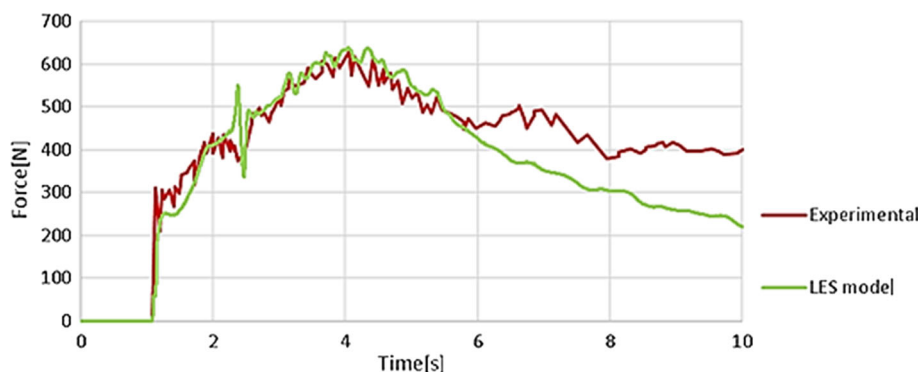


Fig. 11 Test case 4—Time histories of the experimental data. The total hydrodynamic force exerted on a column with square cross section by a dambreak flow considering dry bed condition and without mitigation wall and an initial impundment water depth of 85 cm

5.4 Test Case 4: Effect of tsunami-induced force on a square cross section column without considering mitigation wall, water depth 85 cm

The net force–time histories across the structure surfaces are shown in Fig. 11.

A small spike observed at $t = 1$ s was the result of the initial interaction between the column and the incoming bore. The numerically predicted value was slightly smaller than the experimental one. At approximately $t = 2.5$ s, a distinct rise and subsequent fall in the total force were observed in numerical data for a short period of time which is due to the collapse of the rising volume of water onto the incoming flow. The same trend was also observed in the experimental data but with a smaller magnitude and for a longer period of time. After this instant, numerical and experimental data revealed very good agreement with each other and the run-up force was captured very well by applying the LES model. Similar to the previously calculated force–time histories, the rate of decrease in the calculation of the net force was faster.

5.5 Test Case 5: Effect of tsunami-induced force on a square cross section column with considering inclined (45°) mitigation wall located at 3D

The calculated force imposed on the column (Fig. 12).

Good agreement can be observed between the numerical model results and the laboratory data up to the instant of 5 s. An abrupt initial spike is observed in the experimental data, while the numerical model is not able to capture this spike. The second noticeable rise and fall observed in the physical data are due to the collapse of the rising volume of water, which is less pronounced for the numerical data (Fig. 13).

It can be concluded that positioning the mitigation wall at 3D upstream from the structure is not as effective in reducing the tsunami-induced force on the structure. The resultant peak force is also slightly larger in comparison with the case without wall. The incoming bore overtopped the 45° -inclination wall located at a distance of 3D and was projected into the air and caused the bore front to hit the structure at a higher elevation than in the case without a wall. This phenomenon led to a large impulsive overturning moment.

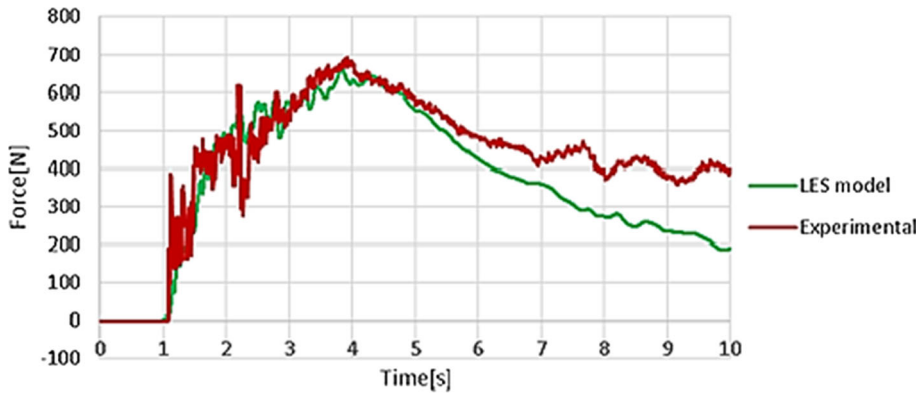


Fig. 12 Test case 5—Total hydrodynamic force by a dambreak flow on a column considering dry bed condition and a mitigation wall located at 3D and an initial impoundment water depth of 85 cm

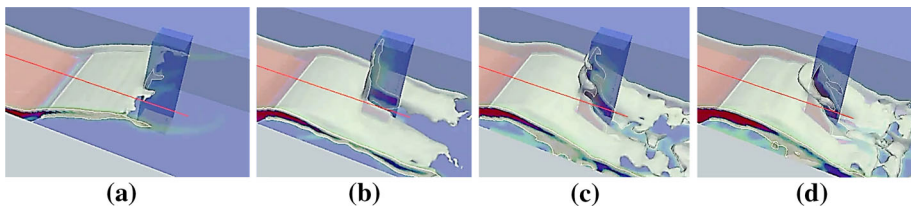


Fig. 13 **a** First interaction of incoming flow and the column. **b** Initial run-up of the impacted bore onto the column. **c** Run-up water starts collapsing onto the incoming flow. **d** Collapse of the rising volume of water onto the incoming flow

6 Conclusions

In this study, the interaction between the forces exerted by tsunami-like bores on free-standing vertical structures in the presence of mitigation walls was simulated using a large eddy simulation model. The configurations investigated included using curved and inclined walls (at 45°), positioned at two different distances from the upstream side of the structure. The problem was studied via applying a three-dimensional hydrodynamic computational model with free surface, RANS turbulence models and an LES model for simulating complex turbulent behavior of the hydrodynamic field. The capabilities of the applied RANS and LES models were compared. It was demonstrated that LES approach with one-equation for subgrid-scale kinetic energy performed better than the employed RANS models such as $k-\epsilon$, $k-\omega$ and LRR . The comparisons were performed using the data obtained from a comprehensive experimental program conducted at the National Research Council (NRC), in Ottawa, Canada. The functionality and effectiveness of the numerical model were validated by comparing the simulated data to experimental results in a wide range of experiments. The numerical model results using the LES model showed good agreement with the measured experimental data. Therefore, LES can be considered as a better choice in modeling the turbulence in such cases. To the best of the authors' knowledge, LES modeling of tsunami bores using an equation for subgrid-scale kinetic energy has not been performed prior to this study. Furthermore, numerical simulation of the effect of low-height mitigation walls on force reduction for tsunami bores has not been conducted

before. The numerical results confirmed that placing mitigation walls at an appropriate distance from the upstream side of the structure can be effective in decreasing the impulsive and run-up forces. The role of the mitigation walls of low height is to decrease the force but not necessarily to prevent overtopping. Numerical results confirmed the experimental findings which indicated that positioning a curved mitigation wall at 3D can lead to a noticeable reduction on exerted net force on the structure, while it is not as effective at 1D. Moreover, placing inclined 45° mitigation wall at 3D was not effective in reducing the net force either due to the “launching” effect which results in the wave hitting the structure at a higher elevation impact point on the structure—this induced a significant increase in the moment of the hydrodynamic force exerted on the structure.

In conclusion, the present study demonstrates that LES with one equation for subgrid-scale kinetic energy is a good approach for simulating turbulence in modeling tsunami bores and confirms that mitigation walls could be effective in reducing the impulsive and run-up forces and can also damp the peak force if placed at an appropriate place.

References

- Al-Faesly T, Palermo D, Nistor I, Cornett A (2012) Experimental modeling of extreme hydrodynamic forces on structural models. *Int J Prot Struct* 3(4):477–505
- Andrillon Y, Alessandrini B (2004) A 2D+ T VOF fully coupled formulation for the calculation of breaking free-surface flow. *J Mar Sci Technol* 8(4):159–168
- Arikawa T (2008, December) Behaviors of concrete walls under impulsive tsunami load. In: AGU fall meeting abstracts, vol 1, p 1331
- Árnason H (2005) Interactions between an incident bore and a free-standing coastal structure. UMI Dissertation Services
- Asakura R, Iwase K, Ikeya T, Takao M, Kaneto T, Fujii N, Ohmori M (2002, July) The tsunami wave force acting on land structures. In: Coastal engineering conference, vol 1, pp 1191–1202. ASCE American Society of Civil Engineers
- Bandara KMK, Dias WPS (2012) Tsunami wave loading on buildings: a simplified approach. *J Nat Sci Found Sri Lanka* 40(3):211–219
- Berberovic E (2010) Investigation of free-surface flow associated with drop impact: numerical simulations and theoretical modeling. Doctoral dissertation, TU Darmstadt/FG Strömungslehre und Aerodynamik
- Chamani MR, Dehghani AA, Beirami MK, Gholipour R (2013) Fluid mechanics, 2nd edn. IUT Publishing, Gazipur, p 209
- Chanson H (2006) Tsunami surges on dry coastal plains: application of dam break wave equations. *Coast Eng J* 48(04):355–370
- Cross RH (1967) Tsunami surge forces. *J Waterw Harb Div* 93(4):201–234
- Cumberbatch E (1960) The impact of a water wedge on a wall. *J Fluid Mech* 7(03):353–374
- Douglas S, Nistor I (2015) On the effect of bed condition on the development of tsunami-induced loading on structures using OpenFOAM. *Nat Hazards* 76(2):1335–1356
- FEMA P646 (2012) Guidelines for design of structures for vertical evacuation from tsunamis. Federal Emergency Management Agency, Washington, DC
- Fujima K, Achmad F, Shigihara Y, Mizutani N (2009) Estimation of tsunami force acting on rectangular structures. *J Disaster Res* 4(6):404–409
- Fukui Y, Nakamura M, Shiraishi H, Sasaki Y (1963) Hydraulic study on tsunami. *Coast Eng Jpn* 6:67–82
- Gotoh H, Shao S, Memita T (2004) SPH-LES model for numerical investigation of wave interaction with partially immersed breakwater. *Coast Eng J* 46(01):39–63
- Gotoh H, Ikari H, Memita T, Sakai T (2005) Lagrangian particle method for simulation of wave overtopping on a vertical seawall. *Coast Eng J* 47(02–03):157–181
- Cox D, Van de Lind J, Wilson J (2010) Proceedings of the International Convention of Society of Wood Science and Technology and United Nations Economic Commission for Europe – Timber Committee
- Heyns JA, Malan AG, Harms TM, Oxtoby OF (2013) Development of a compressive surface capturing formulation for modelling free-surface flow by using the volume-of-fluid approach. *Int J Num Method Fluids* 71(6):788–804

- Huang Y, Zhu C (2015) Numerical analysis of tsunami–structure interaction using a modified MPS method. *Nat Hazards* 75(3):2847–2862
- Kraus NC, McDougal WG (1996) The effects of seawalls on the beach: Part I, an updated literature review. *J Coast Res* 12(3):691–701
- Liu H, Li J, Shao S, Tan SK (2015) SPH modeling of tidal bore scenarios. *Nat Hazards* 75(2):1247–1270
- Lukkunaprasit P, Ruangrassamee A (2008) Building damage in Thailand in the 2004 Indian Ocean tsunami and clues for tsunami-resistant design. *IES J Part A: Civil Struct Eng* 1(1):17–30
- Nouri Y, Nistor I, Palermo D, Cornett A (2010) Experimental investigation of tsunami impact on free standing structures. *Coast Eng J* 52(01):43–70
- OpenFOAM (2014) OpenFOAM: the open source CFD toolbox. <http://www.openfoam.com>. OpenCFD Ltd, OpenFOAM, the Open Source CFD Toolbox, User Guide, OpenCFD Ltd., 2008, <http://www.opencfd.co.uk/openfoam/>
- Oshnack ME, Aguiniga F, Cox D, Gupta R, van de Lindt J (2009) Effectiveness of small onshore seawall in reducing forces induced by Tsunami bore: large scale experimental study. *J Disaster Res* 4(6):382–390
- Piomelli U (1999) Large-eddy simulation: achievements and challenges. *Prog Aerosp Sci* 35(4):335–362
- Rahman S, Akib S, Khan MTR, Shirazi SM (2014) Experimental study on tsunami risk reduction on coastal building fronted by sea wall. *Sci World J*
- Ramsden JD, Raichlen F (1990) Forces on vertical wall caused by incident bores. *J Waterw Port Coast Ocean Eng* 116(5):592–613
- St-Germain P, Nistor I, Townsend R, Shibayama T (2013) Smoothed-particle hydrodynamics numerical modeling of structures impacted by tsunami bores. *J Waterw Port Coast Ocean Eng* 140(1):66–81
- Stoker JJ (1957) *Water waves: the mathematical theory with applications, pure and applied mathematics*, vol. 4. Interscience Publishers, New York
- Thomas S, Cox D (2011) Influence of finite-length seawalls for tsunami loading on coastal structures. *J Waterw Port Coast Ocean Eng* 138(3):203–214
- Thusyanthan NI, Gopal Madabhushi SP (2008, May) Tsunami wave loading on coastal houses: a model approach. In: *Proceedings of the institution of civil engineers-civil engineering*, vol 161, no 2, pp 77–86. Thomas Telford Ltd